Secure-Tunnel: Post-Quantum Cryptographic Overlay for UAV C2 Links

Results and Analysis Section — v1.1 (2026-03-03)

Draft compilation for review

I. Results and Analysis

All measurements were collected on a Raspberry Pi 4 Model B Rev 1.5 (BCM2711, four Cortex-A72 cores at 1.8 GHz, 4 GB RAM) running Debian 12 with kernel 6.12.47 and the `performance` CPU governor. Power was sampled via an INA219 current sensor (0.1 Ω shunt, I^2C bus 1) with a documented V_{BUS} gain correction of 1.22 to compensate for clone-chip ADC drift. Timing used `time.perf_counter_ns()` with garbage collection disabled during timing passes. Board-level idle power measured $P_{\text{idle}} \approx 3.32$ W (30 s idle capture; 3.320 W for AEAD evaluation, 3.309 W for inference benchmarks); reported power values include this baseline unless explicitly stated otherwise. The Cortex-A72 in the RPi4 does *not* implement ARMv8 Crypto Extensions; AES-256-GCM therefore executes in software table-lookup mode, while ChaCha20-Poly1305 benefits from 128-bit NEON SIMD.

Table I defines short aliases used throughout this section. Cryptographic implementations use liboqs-python 0.14.0 (KEM/SIG) and OpenSSL 3.5.4 (AES-256-GCM, ChaCha20-Poly1305). Ascon-128a uses a native C binding (ascon-c v1.2, `opt64`) via `ctypes`.

TABLE I
Algorithm Aliases and NIST Security Levels

Alias	Full Name	Level	Type
ML512	ML-KEM-512	L1	KEM
ML768	ML-KEM-768	L3	KEM
ML1024	ML-KEM-1024	L5	KEM
HQ128	HQC-128	L1	KEM
HQ192	HQC-192	L3	KEM
HQ256	HQC-256	L5	KEM
MC348	Classic-McEliece-348864	L1	KEM
MC460	Classic-McEliece-460896	L3	KEM
MC8192	Classic-McEliece-8192128	L5	KEM
DSA44	ML-DSA-44	L1	SIG
DSA65	ML-DSA-65	L3	SIG
DSA87	ML-DSA-87	L5	SIG
F512	Falcon-512	L1	SIG
F1024	Falcon-1024	L5	SIG
SP128	SPHINCS$^+$-SHA2-128s	L1	SIG
SP192	SPHINCS$^+$-SHA2-192s	L3	SIG
SP256	SPHINCS$^+$-SHA2-256s	L5	SIG
AESG	AES-256-GCM	—	AEAD
CH20	ChaCha20-Poly1305	—	AEAD
ASC	Ascon-128a	—	AEAD

A. KEM Primitive Evaluation

Table II presents timing and energy measurements for all nine KEM primitives, each measured over 100 iterations with concurrent INA219 power sampling at 1 kHz.

ML-KEM dominates across all security levels, completing a full keygen-encapsulate-decapsulate cycle in under 0.72 ms at Level 5. HQC's code-based design incurs 255–1 066× higher total cost than ML-KEM at equivalent levels, driven primarily by decapsulation. Classic McEliece exhibits extreme keygen asymmetry: MC8192 keygen requires 9.02 s (39.75 J) but encapsulation completes in 1.89 ms, making it practical only in scenarios where key pairs are pre-provisioned. Power draw during KEM operations ranges from 3.44 W (ML512 keygen) to 4.40 W (MC8192 keygen), a 28% spread indicating CPU-bound computational intensity for the heavier algorithms.

B. Signature Primitive Evaluation

Table III evaluates eight signature schemes. Sign and verify are the per-handshake operations; keygen is a one-time cost.

Falcon achieves the lowest combined sign+verify latency at both L1 (0.970 ms) and L5 (1.732 ms), outperforming ML-DSA by 36% and 33% respectively, though its keygen is 51–73× slower (a one-time cost amortized over the key lifetime). SPHINCS$^+$, the only hash-based scheme, imposes prohibitive signing latency: SP128 at 1.47 s and SP192 at 2.61 s, consuming 6.01 J and 10.63 J per signature. For UAV handshakes occurring at rekey intervals of 10–120 s, SPHINCS$^+$ signing alone can approach the rekey period, making it viable only for conservative fallback configurations. Verification remains fast for all algorithms ($\leq$3.3 ms), critical for the GCS which must verify each incoming handshake.

C. AEAD Primitive Evaluation

Data-plane encryption uses one of three AEAD ciphers. AEAD selection does *not* affect handshake timing; it operates exclusively on the UDP data plane after key establishment. Table IV reports results from 10 000 iterations per configuration with a 500-iteration warmup, using a two-pass methodology (timing pass with GC disabled, power pass with background

TABLE II
KEM PRIMITIVE PERFORMANCE ON RPI4 @ 1.8 GHZ (100 ITERATIONS, 1 KHZ INA219)

Alias	**Level**	**Finalist**	**Keygen** (ms)	**Encaps** (ms)	**Decaps** (ms)	$P_{\mathbf{avg}}$ (W)	$E_{\mathbf{enc}}$ (μJ)	$E_{\mathbf{dec}}$ (μJ)	ρ
NIST Level 1									
ML512	L1	✓	0.240	0.152	0.158	3.46	526	551	1.00×
HQ128	L1		22.33	44.68	73.39	3.78	167 590	284 510	255×
MC348	L1		351.0	0.466	55.98	3.82	1 619	208 294	741×
NIST Level 3									
ML768	L3	✓	0.237	0.182	0.185	3.49	636	653	1.00×
HQ192	L3		67.68	135.8	211.9	3.94	537 757	856 128	688×
MC460	L3		1 281	0.872	89.59	3.92	3 135	339 956	2 271×
NIST Level 5									
ML1024	L5	✓	0.273	0.216	0.231	3.48	754	806	1.00×
HQ256	L5		124.4	250.2	392.6	4.06	1 023 984	1 620 633	1 066×
MC8192	L5		9 021	1.894	210.1	4.00	6 849	829 782	12 823×

P_{avg}: mean of keygen/encaps/decaps power readings. E_{enc}/E_{dec}: energy per encapsulate/decapsulate. ρ: total KEM cost relative to ML-KEM at same NIST level. Board-level power includes idle draw ($P_{\text{idle}} \approx 3.32\,\text{W}$).

TABLE III
SIGNATURE PRIMITIVE PERFORMANCE ON RPI4 @ 1.8 GHZ (100 ITERATIONS, 1 KHZ INA219)

Alias	**Level**	**Finalist**	**Keygen** (ms)	**Sign** (ms)	**Verify** (ms)	$P_{\mathbf{avg}}$ (W)	$E_{\mathbf{sign}}$ (μJ)	$E_{\mathbf{verify}}$ (μJ)	ρ
NIST Level 1									
DSA44	L1	✓	0.375	1.179	0.331	3.56	4 264	1 175	1.00×
F512	L1	✓	19.26	0.778	0.192	3.58	2 773	683	0.64×
SP128	L1		194.2	1 468	1.582	3.91	6 008 719	5 690	972×
NIST Level 3									
DSA65	L3	✓	0.557	1.633	0.484	3.56	5 849	1 723	1.00×
SP192	L3		282.6	2 610	2.299	3.93	10 628 558	8 318	1 234×
NIST Level 5									
DSA87	L5	✓	0.748	1.847	0.723	3.55	6 612	2 564	1.00×
F1024	L5	✓	54.24	1.452	0.280	3.62	5 175	995	0.67×
SP256	L5		187.0	2 321	3.292	3.90	9 516 164	11 738	904×

ρ: sign+verify cost relative to ML-DSA at same level. Falcon $\rho < 1$ reflects faster combined sign+verify. SP128 sign = 1.47 s (6.01 J) per invocation.

TABLE IV
AEAD PERFORMANCE AT 256-BYTE PAYLOAD (10K ITER, RPI4, RERUN)

Alias	**Enc** (μs)	**Dec** (μs)	$P_{\mathbf{enc}}$ (W)	$P_{\mathbf{dec}}$ (W)	$E_{\mathbf{256B}}$ (μJ)	$E_{\mathbf{bit}}$ (nJ/bit)
AESG	9.25	9.09	3.63	3.57	66.0	32.2
CH20	6.47	6.05	3.54	3.58	44.6	21.8
ASC	12.72	12.85	3.84	3.59	95.0	46.4

$E_{256\text{B}} = P_{\text{avg}} \times (t_{\text{enc}} + t_{\text{dec}})$. $E_{\text{bit}} = E_{256\text{B}}/(256 \times 8)$. Board-level power; $P_{\text{idle}} = 3.320\,\text{W}$ not subtracted. Crypto-only ΔP: AESG 0.31, CH20 0.22, ASC 0.52 W above idle.

TABLE V
AEAD ENCRYPT LATENCY SCALING BY PAYLOAD SIZE (μS, RERUN)

Alias	**9 B**	**33 B**	**64 B**	**256 B**	**1024 B**	**4096 B**
AESG	4.75	5.45	5.96	9.25	22.98	75.71
CH20	5.30	5.38	5.33	6.47	9.38	19.25
ASC	11.38	11.82	11.78	12.72	17.27	30.86

Setup overhead dominates at small payloads. CH20 scales at 3.4 μs/KiB; AESG at 17.7 μs/KiB; ASC at 4.8 μs/KiB.

INA219 sampling at ~103 Hz). The Ascon-128a backend was verified as native C (ascon-c v1.2, opt64, `ctypes` binding loading `libascon128a.so`), passing the AEAD known-answer test at runtime.

ChaCha20-Poly1305 is the fastest AEAD on this platform at 6.47 μs for encryption, 30% faster than AES-256-GCM (9.25 μs) despite AES being generally considered the hardware-accelerated standard. The inversion is explained by the absence of ARMv8 Crypto Extensions on the Cortex-A72: AES-GCM falls back to software table-lookup while ChaCha20 exploits 128-bit NEON SIMD [rfc8439]. Ascon-128a, using a native C binding (ascon-c v1.2 `opt64` via `ctypes`), runs 1.97× slower than CH20 and 1.37× slower than AESG.

Table V shows AEAD scaling across payload sizes typical of MAVLink telemetry (9–263 bytes) and bulk transfer (1024–4096 bytes).

At the minimum MAVLink heartbeat size (9 bytes), per-packet overhead is 4.75 μs (AESG), 5.30 μs (CH20), and 11.38 μs (ASC). CH20 and AESG converge below 64 bytes because setup cost dominates; beyond 256 bytes, CH20's superior NEON throughput yields a widening gap. At 4096 bytes, the ratio CH20:AESG reaches 1:3.93. Ascon-128a's flat initialization cost (native C per-call `ctypes` overhead) makes it competitive only below 64 bytes.

The energy-per-bit metric E_{bit} at the 256-byte reference payload yields CH20 = 21.8 nJ/bit, AESG = 32.2 nJ/bit, and ASC = 46.4 nJ/bit. For a typical MAVLink stream at 50 Hz with 263-byte packets (the default MAVLink v2 maximum), the per-second AEAD energy cost is 2.16 mJ (CH20), 3.42 mJ (AESG), or 4.47 mJ (ASC) — negligible relative to the 3.32 W board idle draw.

D. Suite-Level Evaluation

The system supports $9 \times 8 \times 3 = 216$ theoretical suite combinations, of which 72 are level-matched. Since AEAD does not affect handshake timing, the crypto-only handshake cost is determined solely by the KEM and SIG components:

$$T_{\text{HS}} = t_{\text{keygen}} + t_{\text{encaps}} + t_{\text{decaps}} + t_{\text{sign}} + t_{\text{verify}} \tag{1}$$

Table VI presents representative suites per NIST level, with T_{HS} reconstructed from primitive benchmarks to isolate cryptographic cost from network latency.

The default suite `cs-mlkem768-aesgcm-mldsa65` achieves a 2.72 ms crypto-only handshake cost at Level 3, consuming 9.6 mJ. At Level 5, ML1024+F1024 requires only 2.45 ms—faster than the Level 3 default because Falcon-1024 verify (0.280 ms) is 42% faster than DSA65 verify (0.484 ms).

The dominant cost factor varies by KEM family: for ML-KEM suites, the signature accounts for 73–99% of T_{HS}; for HQC and McEliece suites, the KEM accounts for 98–99%. SPHINCS$^+$ suites exhibit $T_{\text{HS}} > 1$ s regardless of KEM choice, restricting them to pre-provisioned or infrequent-rekey scenarios.

E. DDoS Impact on PQC Performance

Concurrent DDoS detection imposes computational overhead that competes with the cryptographic pipeline for CPU, memory, and thermal headroom. Table VII quantifies the system-level impact of activating XGBoost and TST (Transformer-based[1]) detectors.

XGBoost raises CPU utilization from 1.2% to 55.3% and adds 505 mW (+13.9%) to board power. TST saturates the CPU at 93.0%, adding 690 mW (+19.0%) and raising temperature by 11.3°C. This thermal increase is critical: the Cortex-A72 throttles at 80°C, leaving only 16.2°C of headroom under TST versus 27.5°C without detection.

Table VIII shows the per-packet AEAD latency impact when a detector runs concurrently with the encrypted tunnel.

XGBoost adds approximately 32% to per-packet AEAD latency due to CPU contention on the shared 4-core processor. TST shows a heterogeneous impact: CH20 degrades least (+14.4%) thanks to its NEON-optimized path requiring fewer CPU stalls, while ASC degrades most (+41.9%) due to its longer per-packet duration competing for shared execution resources. The scheduler's MDEAS policy addresses this interaction by forbidding Ascon-128a (the slowest AEAD) when TST is active, and by discouraging McEliece KEMs under TST due to their extended CPU-bound keygen.

[1]TST uses `seq_len=1`, reducing the self-attention mechanism to an identity operation; the model is functionally a 3-layer MLP with 1.6 M parameters.

F. PQC Impact on DDoS Detection

The converse question is how PQC workload affects detector efficacy. Table IX compares the four trained detectors on inference performance.

XGBoost achieves the highest reported accuracy (94.55%) with competitive latency (7.42 ms mean, 9.52 ms P99) and moderate power overhead (+951 mW). LightGBM offers the lowest latency (6.72 ms) and energy (29.0 mJ) at a marginal accuracy reduction (93.47%). RandomForest is an outlier with 74.68 ms mean inference latency and 291.4 mJ energy per inference—10× the energy of XGBoost—due to its 200-tree ensemble with max_depth=15 and 674 MB RAM footprint. TST consumes the most power (+1 973 mW) at 93% CPU, delivering the lowest accuracy (90.27%) among the four models.

For the deployed two-tier cascade (XGBoost primary, TST secondary), the combined power overhead is bounded by the active detector at any given time, as the scheduler runs at most one detector subprocess concurrently.

G. Scheduler Behavioral Impact

The Multi-Dimensional Energy-Aware Scheduler (MDEAS) operates three independent decision axes evaluated at 1 Hz:

1) **AEAD axis:** selects among AESG, CH20, ASC based on energy-per-bit and thermal state;
2) **Security-level axis:** promotes/demotes between L1/L3/L5 suites, triggering a full PQC rekey;
3) **DDoS-detector axis:** activates/deactivates XGBoost or TST based on threat telemetry and thermal headroom.

1) Rekey Overhead Model: Each security-level transition requires a complete 3-round PQC handshake. The rekey overhead fraction is:

$$\Phi(R) = \frac{T_{\text{HS}}}{R + T_{\text{HS}}} \tag{2}$$

where R is the rekey interval and T_{HS} is the handshake crypto cost from (1). For the default suite (ML768+DSA65, $T_{\text{HS}} = 2.72$ ms) at the minimum break-even interval of 30 s: $\Phi(30) = 2.72 \times 10^{-3}/30.003 = 9.1 \times 10^{-5}$ (<0.01% overhead). Even the heaviest suite (MC8192+SP256, $T_{\text{HS}} \approx 11.6$ s) at 30 s interval yields $\Phi = 0.279$ (27.9% overhead), which the break-even calculator correctly identifies as non-viable.

2) Cost Model: The MDEAS cost function for AEAD selection is:

$$C = w_1 \cdot E_{\text{bit}} + w_2 \cdot T_{\text{HS}} + w_3 \cdot \Delta T + w_4 \cdot D \tag{3}$$

where E_{bit} is the energy per bit (nJ/bit), T_{HS} is the handshake cost (ms), ΔT is the temperature above baseline (°C), and D is the DDoS threat level. The scheduler seeds cost estimates from Table X and refines them via exponential weighted moving average (EWMA, $\alpha = 0.2$ after 10 initial samples at $\alpha = 0.5$).

TABLE VI
REPRESENTATIVE SUITE HANDSHAKE CRYPTO COST (RECONSTRUCTED FROM PRIMITIVES)

KEM	**SIG**	**Level**	t_{KEM} (ms)	t_{SIG} (ms)	T_{HS} (ms)	E_{HS} (mJ)	ρ
NIST Level 1 — Fastest							
ML512	F512	L1	0.550	0.970	1.52	5.3	1.00×
ML512	DSA44	L1	0.550	1.510	2.06	7.3	1.36×
HQ128	F512	L1	140.4	0.970	141.4	528	93.0×
HQ128	DSA44	L1	140.4	1.510	141.9	530	93.4×
MC348	F512	L1	407.4	0.970	408.4	1 579	269×
ML512	SP128	L1	0.550	1 470	1 470	5 737	967×
NIST Level 3 — Default (⋆)							
ML768	DSA65	L3	0.604	2.117	2.72	9.6	1.00×
HQ192	DSA65	L3	415.3	2.117	417.4	1 620	153×
MC460	DSA65	L3	1 372	2.117	1 374	5 358	505×
ML768	SP192	L3	0.604	2 612	2 613	10 290	961×
NIST Level 5 — Maximum							
ML1024	F1024	L5	0.720	1.732	2.45	8.6	1.00×
ML1024	DSA87	L5	0.720	2.570	3.29	11.6	1.34×
HQ256	F1024	L5	767.2	1.732	768.9	3 109	314×
MC8192	DSA87	L5	9 233	2.570	9 235	36 800	3 770×
ML1024	SP256	L5	0.720	2 324	2 325	9 105	949×

$t_{\text{KEM}} = t_{\text{keygen}} + t_{\text{encaps}} + t_{\text{decaps}}$. $t_{\text{SIG}} = t_{\text{sign}} + t_{\text{verify}}$. E_{HS} estimated as $P_{\text{avg}} \times T_{\text{HS}}$. ρ: relative to fastest suite at same NIST level. (⋆) System default: `cs-mlkem768-aesgcm-mldsa65`.

TABLE VII
SYSTEM RESOURCE IMPACT OF DDOS DETECTORS

Phase	**CPU** (%)	**Temp** (°C)	**Power** (mW)	ΔP (%)	**RAM** (MB)
Baseline	1.2	52.5	3 630	—	212
+ XGBoost	55.3	61.5	4 135	+13.9	338
+ TST	93.0	63.8	4 320	+19.0	475

Baseline: PQC tunnel active, no detector. Power from INA219 board-level measurement. CPU: average across 4 cores.

TABLE VIII
AEAD PER-PACKET LATENCY UNDER DDOS DETECTION (NS)

AEAD	**Baseline**	**+ XGB**	**Δ%**	**+ TST**	**Δ%**
AESG	78 045	103 232	+32.3	95 059	+21.8
CH20	63 500	84 327	+32.8	89 407	+14.4
ASC	1 327 100	1 714 654	+29.5	1 878 849	+41.9

Data from DDoS_PQC_IMPACT_ANALYSIS.md 5-phase measurement. Note: these measurements use the scheduler's seed-timing context (see Table X), not the 10k-iteration micro-benchmark. ASC baseline (1.33 ms) reflects scheduler seed values; micro-benchmark ASC = 12.72 μs. Ascon-128a is forbidden under TST by scheduler constraint.

3) *Cross-Axis Constraints:* The MDEAS enforces interaction constraints between axes to prevent pathological configurations:

- Ascon-128a is *forbidden* when TST is active, as its 1.33 ms per-packet cost combined with 93% CPU utilization exceeds the real-time tunnel budget.
- McEliece KEMs (MC348/MC460/MC8192) are *discouraged* under TST due to CPU-bound keygen competing with the detector.
- McEliece × SPHINCS$^+$ cross-level pairs are forbidden under all detector states, as T_{HS} would exceed the 45 s rekey timeout (`REKEY_HANDSHAKE_TIMEOUT`).
- The minimum break-even interval is 30 s (`_MIN_BREAK_EVEN_S`), preventing oscillatory switching.
- Detector transitions include warmup durations: XGBoost 10 s, TST 5 s.

TABLE IX
DDOS DETECTOR INFERENCE ON RPI4 (REPORTED METRICS)

Model	**Acc.** (%)	**F1**	t_{inf} (ms)	**P99** (ms)	ΔP (mW)	E_{inf} (mJ)
LightGBM	93.47†	0.937	6.72	8.59	+999	29.0
XGBoost	94.55†	0.947	7.42	9.52	+951	31.7
RF	93.35†	0.935	74.68	102.63	+594	291.4
TST	90.27†	0.80	10.72	26.12	+1 973	56.7

†Accuracy values are as reported in model documentation; no independent evaluation scripts or confusion matrices were available for verification. All models trained on CIC-IoT-2023 (3.4M samples). XGBoost/LightGBM/RF: 54 features, 15 classes. TST: 46 features, 6 classes.

TABLE X
MDEAS AEAD COST SEEDS (FROM `POLICY.PY`)

AEAD	t_{enc} (μs)	t_{dec} (μs)	P (W)	ΔT (°C)
AESG	66.9	73.6	3.595	−0.1
CH20	63.5	70.7	3.559	0.0
ASC	1 327	961	3.558	+1.5

Seed values from `individual_benchmarks/raw_data/5iter_test` and live suite runs (2026-02-19). Note: seed timing uses a different measurement context (5 iterations, `ondemand` governor) than the 10k-iteration benchmark data; the EWMA converges to production values within the first 10 observations.

TABLE XI
MDEAS DETECTOR OVERHEAD PARAMETERS

Detector	ΔP (W)	ΔT (°C)	**ΔCPU** (pp)	$T_{\mathbf{max}}$ (°C)
None	0.00	0.0	0	80.0
XGBoost	0.95	4.8	35	75.0
TST	1.97	10.7	91	69.0

$T_{\max}$: maximum baseline temperature before detector activation is blocked (80°C throttle point minus ΔT). ΔCPU in percentage points. Values from `sscheduler/policy.py` lines 683–695.

Table XI summarizes the per-detector resource deltas used by the scheduler for proactive thermal management.

The interplay between cryptographic cost and detection overhead creates a three-dimensional Pareto surface: increasing security level raises T_{HS} and E_{HS}, enabling detection raises ΔP and ΔT, and selecting a heavier AEAD raises E_{bit} and ΔT. The MDEAS navigates this surface at 1 Hz using telemetry feedback, ensuring the system remains within the thermal envelope while maintaining the configured minimum security level.

H. Cross-Dimensional Tradeoff Analysis

The preceding sections evaluate KEM, SIG, AEAD, DDoS, and scheduler dimensions independently. This section presents three cross-cutting comparison tables that expose the interactions between security posture, symmetric cipher choice, adversarial load, and scheduler behavior.

1) Security-Cost Tradeoff Under Adversarial Load: Table XII combines handshake cost (from Table VI) with detector overhead (from Table XI) to show the full operational cost of representative security configurations.

The ML-KEM + Falcon/ML-DSA suites at all three NIST levels maintain sub-3.3 ms handshake times, leaving ample margin for concurrent DDoS detection. The cost gap between best (ML-KEM) and worst (McEliece/HQC) suites ranges from 93× at L1 to 3 770× at L5, establishing ML-KEM as the only viable family for latency-constrained UAV rekey.

2) AEAD Sensitivity to DDoS Load [LAYOUT]*:* Table XIII quantifies the per-packet AEAD latency degradation under concurrent DDoS detection, combining the micro-benchmark baseline from Table IV with the cross-impact measurements.

Two patterns emerge from the provisional data: (1) XGBoost imposes a roughly uniform ∼32% latency penalty across all three AEAD ciphers, consistent with its measured 35 pp CPU consumption leaving reduced core availability; (2) TST affects the ciphers asymmetrically—CH20 degrades least (+14.4%) while ASC degrades most (+41.9%)—because NEON-pipelined instructions interleave more efficiently with the Transformer inference workload on the out-of-order Cortex-A72 than scalar C or table-lookup AES code. These findings will be refined when the 54-feature detector infrastructure is finalized.

3) Scheduler Decision Trace [LAYOUT]*:* Table XIV illustrates the MDEAS state-machine response to five representative operational scenarios, derived from the constraint rules and thresholds in `sscheduler/policy.py`. No measured runtime decision logs are available (see inconsistency report I4); all entries are architectural design points.

The trace illustrates three key scheduler behaviors: (1) thermal self-protection via detector deactivation when headroom is exhausted (Scenario 2); (2) automatic AEAD switch enforcement when cross-axis constraints are violated (Scenario 4, ASC forbidden with TST); and (3) graceful security escalation that avoids pathological KEM×detector combinations (Scenario 5, McEliece KEMs discouraged under TST). These architectural decision points will be validated against measured runtime traces once the scheduler integration tests are complete.

TABLE XII
SECURITY-COST TRADEOFF: HANDSHAKE + DETECTOR OVERHEAD AT REPRESENTATIVE SUITES

Level	**Suite**	ρ	$T_{\mathbf{HS}}$ (ms)	$E_{\mathbf{HS}}$ (mJ)	**System $P_{\mathbf{total}}$ (W)** None	+XGB	+TST	$T_{\mathbf{head}}$ (°C)	**Viable?**
NIST Level 1									
L1	ML512+F512	1.0×	1.52	5.3	3.45	4.40	5.42	27.5/22.7/16.8	✓
L1	HQ128+DSA44	93.4×	141.9	530	3.45	4.40	5.42	27.5/22.7/16.8	✓⋆
NIST Level 3 — System Default									
L3	ML768+DSA65	1.0×	2.72	9.6	3.45	4.40	5.42	27.5/22.7/16.8	✓
L3	HQ192+DSA65	153×	417.4	1 620	3.45	4.40	5.42	27.5/22.7/16.8	✓⋆
NIST Level 5									
L5	ML1024+F1024	1.0×	2.45	8.6	3.45	4.40	5.42	27.5/22.7/16.8	✓
L5	MC8192+DSA87	3 770×	9 235	36 800	3.45	4.40	5.42	20.0/15.2/—	×

P_{total}: $P_{\text{idle}} + \Delta P_{\text{detector}}$ (crypto overhead is negligible at board level). T_{head}: thermal headroom to 80°C throttle point (None/XGB/TST), from baseline 52.5°C. ⋆HQ suites viable but $T_{\text{HS}} > 100$ ms limits rekey frequency. ×MC8192+TST: T_{HS} = 9.2 s with TST at 93% CPU is non-viable (exceeds 45 s timeout under load). ρ: relative to fastest suite at same NIST level.

TABLE XIII
AEAD×DDOS INTERACTION MATRIX AT 256 B [LAYOUT]

AEAD	**Baseline** t (μs)	E_{bit} (nJ/b)	**+ XGBoost** t (μs)	$\Delta\%$	**+ TST** t (μs)	$\Delta\%$
AESG	78.0	—	103.2	+32.3	95.1	+21.8
CH20	63.5	—	84.3	+32.8	89.4	+14.4
ASC	1327	—	1715	+29.5	1879	+41.9

[LAYOUT] — DDoS subsystem under active development. Values from DDoS_PQC_IMPACT_ANALYSIS.md seed-context runs (not micro-benchmark); see inconsistency report I5. Gray cells indicate provisional data awaiting re-benchmark with finalized 54-feature detectors. E_{bit} column pending per-packet power under load. ASC values reflect scheduler seed context (∼1.3 ms), not the native C micro-benchmark (12.7 μs).

TABLE XIV
MDEAS SCHEDULER DECISION TRACE — ARCHITECTURAL SCENARIOS [LAYOUT]

#	Scenario	Trigger	AEAD	Level	Detector	Constraint / Rationale
1	Cold start	$T < 55°$C, no threat	CH20	L3	None	Default suite: lowest E_{bit} at L3
2	Thermal stress	$T > 70°$C sustained	CH20 $\rightarrow$ CH20	L3 $\rightarrow$ L1	XGB $\rightarrow$ None	XGB blocked (T_{max}=75°C headroom violated); level demotion reduces E_{HS}
3	DDoS alert	Threat telemetry $> \theta$	CH20	L3	None $\rightarrow$ XGB	Primary detector activated; 10 s warmup; ASC forbidden if TST escalation
4	Sustained load	$P > 4.5$ W, CPU $> 80\%$	ASC $\rightarrow$ CH20	L3	TST $\rightarrow$ XGB	ASC forbidden with TST; downgrade detector to reduce ΔP by 1.02 W
5	Security escalation	External L5 mandate	CH20	L3 $\rightarrow$ L5	XGB	ML1024+F1024 rekey (T_{HS}=2.45 ms); McEliece forbidden with TST

[LAYOUT] — Scheduler under active development; no runtime decision logs available. Scenarios derived from `sscheduler/policy.py` constraint rules (lines 704–726) and `_DETECTOR_OVERHEAD` parameters (lines 683–695). Gray cells indicate design-time values, not measured transitions. θ: threat detection threshold (configurable).